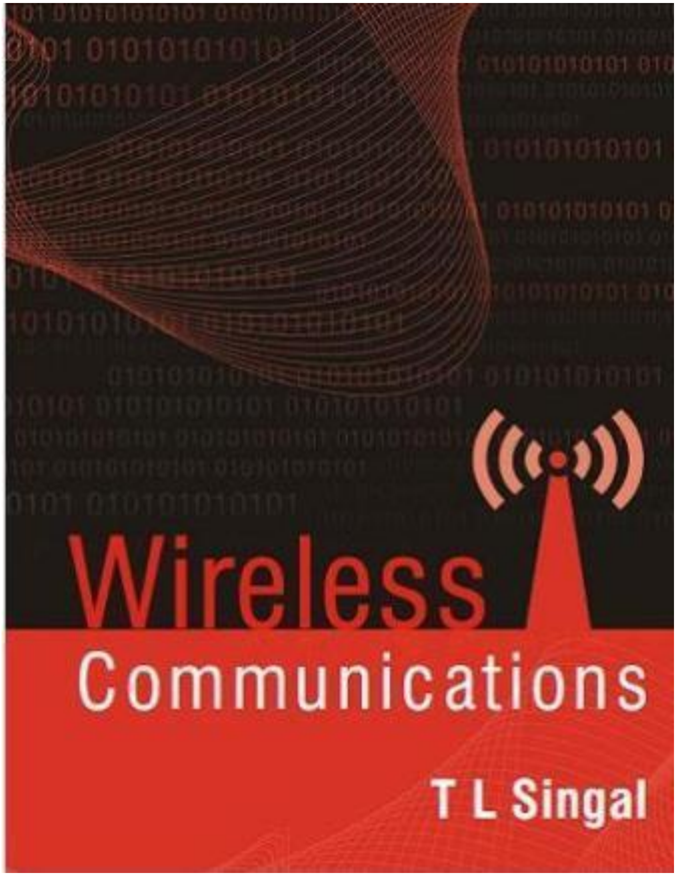
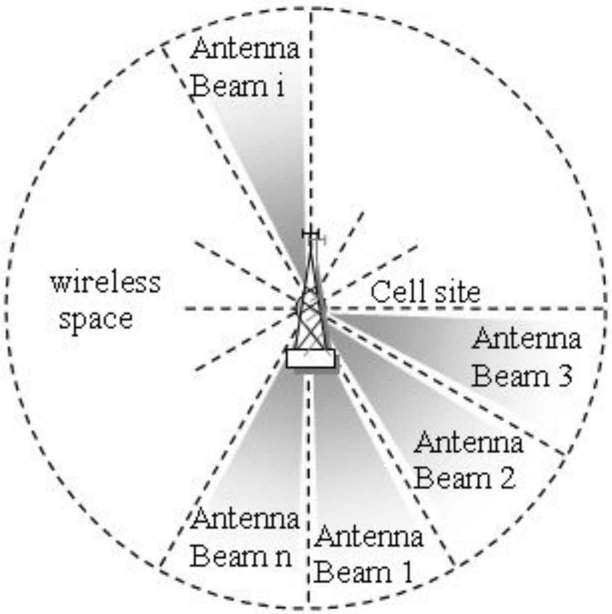
Education

PowerPoint Slides



Multiple Access Techniques



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Multiple Access Techniques (1)

- Introduction
- Frequency Division Multiple Access
- Time Division Multiple Access
- Spread Spectrum Multiple Access
- Space Division Multiple Access

Multiple Access Techniques (2)

- Hybrid Multiple Access Techniques
- Comparison of Multiple Access Techniques
- Packet Radio Multiple Access Techniques
- Carrier Sense Multiple Access Protocols
- Multicarrier Multiple Access Schemes

Why is Multiple Access needed?

- To share the physical resources of a wireless medium by a large number of mobile subscribers at the same time in different locations
- To enables many mobile users to establish communication links simultaneously within the allocated radio spectrum
- To achieve high capacity while maintaining acceptable quality of communication
- To maximize the spectrum utilization

Two types of duplex systems are utilized: frequency division duplexing (FDD) divides the frequency used, and

time division duplexing (TDD) divides the same frequency by time.

FDMA mainly uses FDD,
while TDMA and CDMA systems use either FDD or TDD

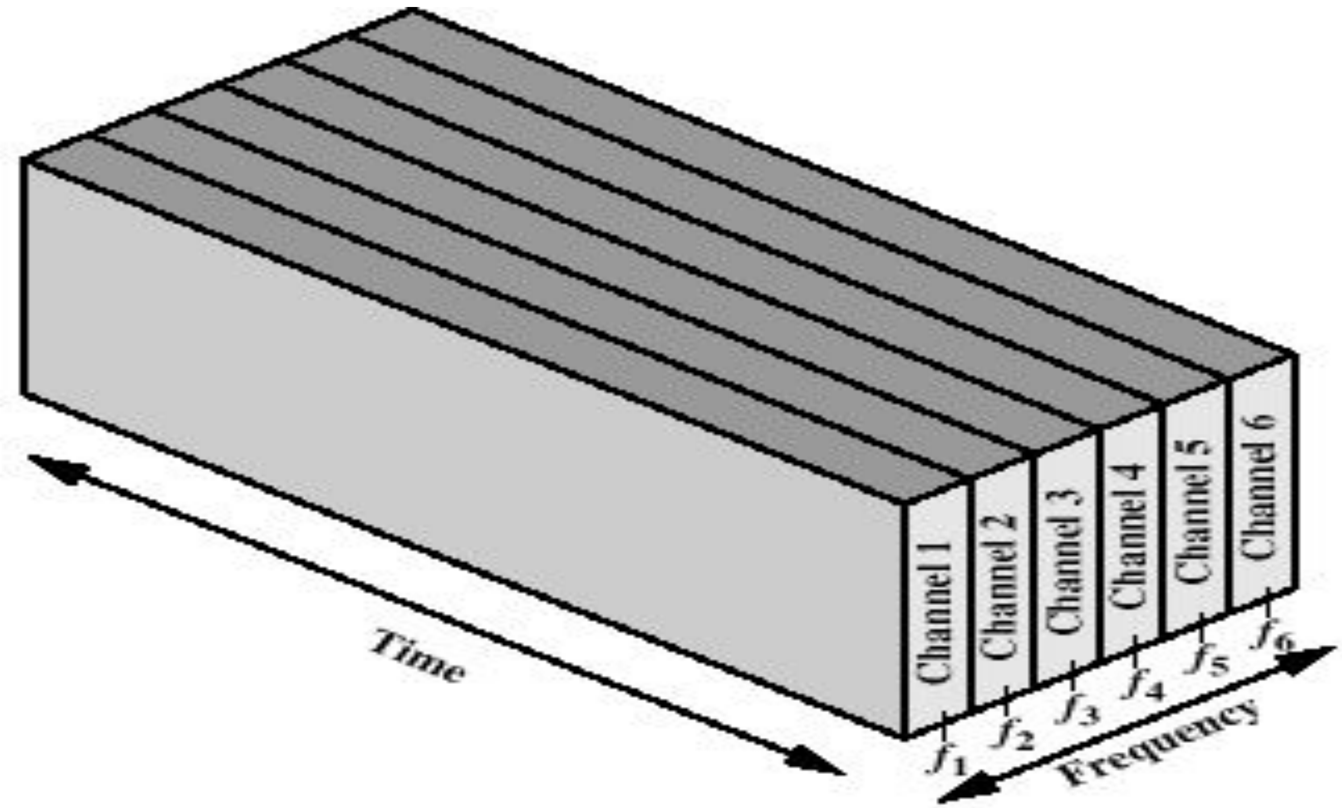
Table 1: MA techniques in different wireless communication systems

Advanced Mobile Phone Systems:	FDMA/FDD
Global System for Mobile:	TDMA/FDD
U.S. Digital Cellular:	TDMA/FDD
Japanese Digital Cellular:	TDMA/FDD
CT2 Cordless Telephone:	FDMA/TDD
Digital European Cordless Telephone:	FDMA/TDD
U.S. Narrowband Spread Spectrum (IS-95):	CDMA/FDD

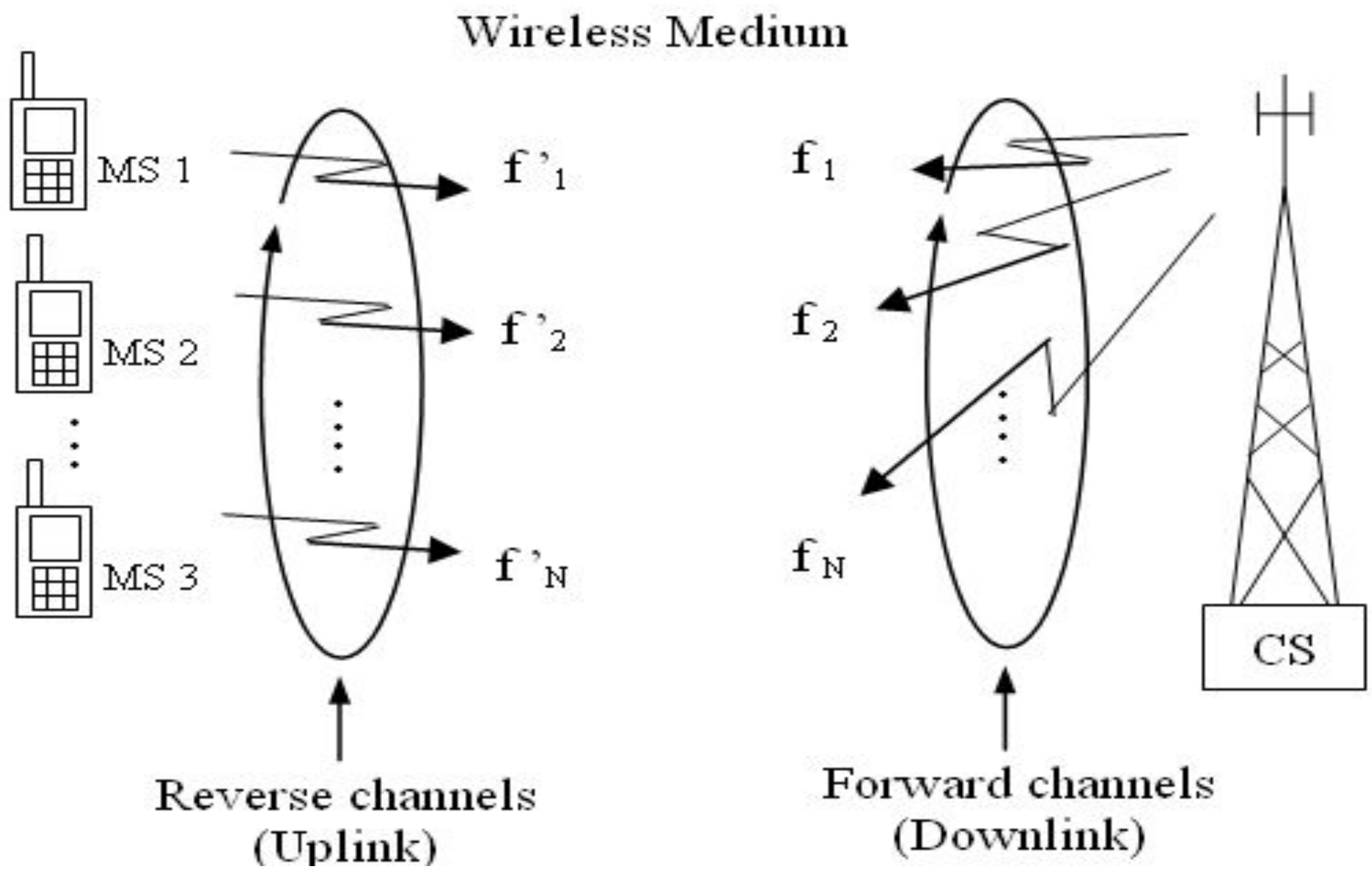
What is FDMA?

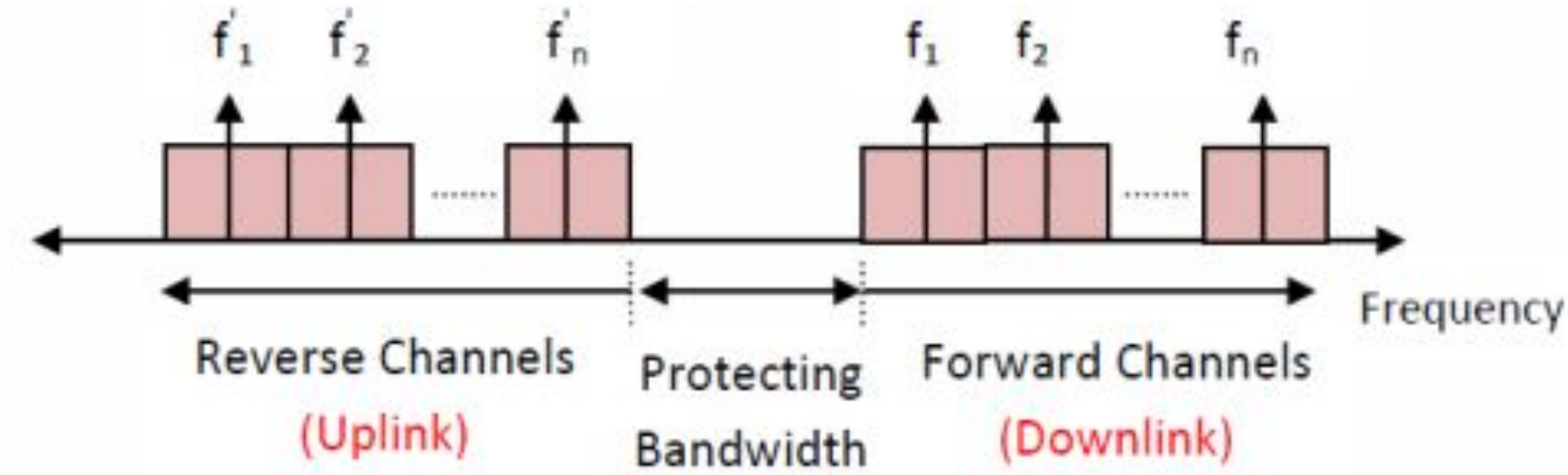
- FDMA refers to sharing the available radio spectrum by assigning specific frequency channels to subscribers either on a permanent basis or on a temporary basis.
- Dividing the whole spectrum into smaller frequency bands
- A channel gets a certain band of the spectrum for whole time

Frequency Division Multiple Access (FDMA)

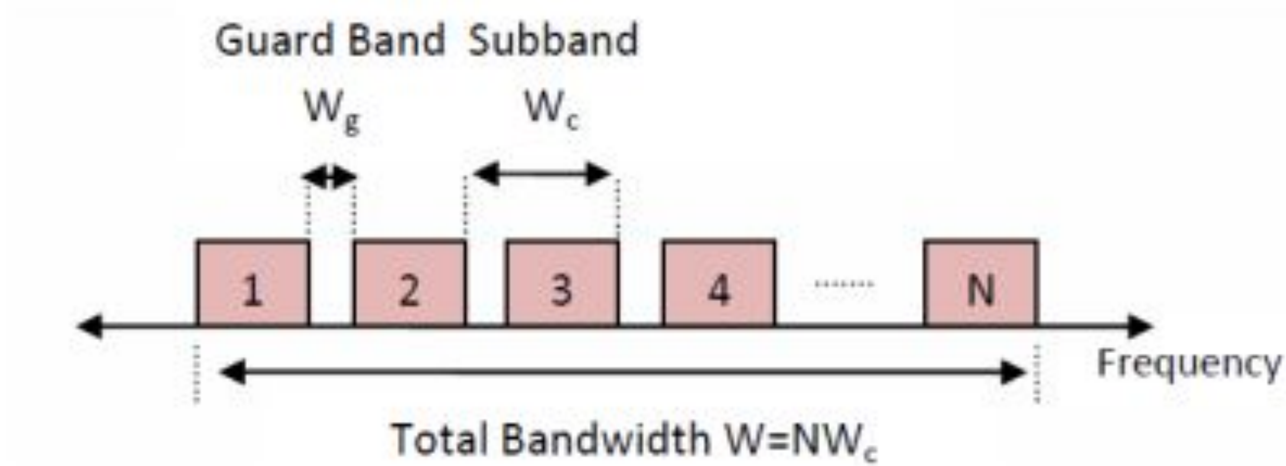


Basic Structure of FDMA System





Structure of forward and reverse channels in FDMA



Guard Band in FDMA

The features of FDMA are as follows:

1. The FDMA channel carries only one phone circuit at a time. If an FDMA channel is not in use, then it sits idle and it cannot be used by other users to increase share capacity. After the assignment of the voice channel the BS and the MS transmit simultaneously and continuously.
2. The bandwidths of FDMA systems are generally narrow
3. The data rate is very less.
4. The complexity of the FDMA mobile systems is lower than that of TDMA mobile systems.
5. FDMA requires tight filtering to minimize the adjacent channel interference.

Advantages:

1. If the channel is not in use, its sits idle.
2. Channel Bandwidth is narrow (30KHz)
3. Fairly efficient when the number of stations is small and the traffic is uniformly constant
4. Uses existing hardware and hence this technology is cost efficient
5. Network timing is not required, hence making the system less complex.

Disadvantages

1. The presence of guard bands.
2. Requires right RF filtering to minimize adjacent channel interference
3. Maximum bit rate per channel is fixed.
4. Small inhibiting flexibility in bit rate capability.
5. Flexibility in channel allocation is less.
6. Uplink power control is required to maintain the link quality

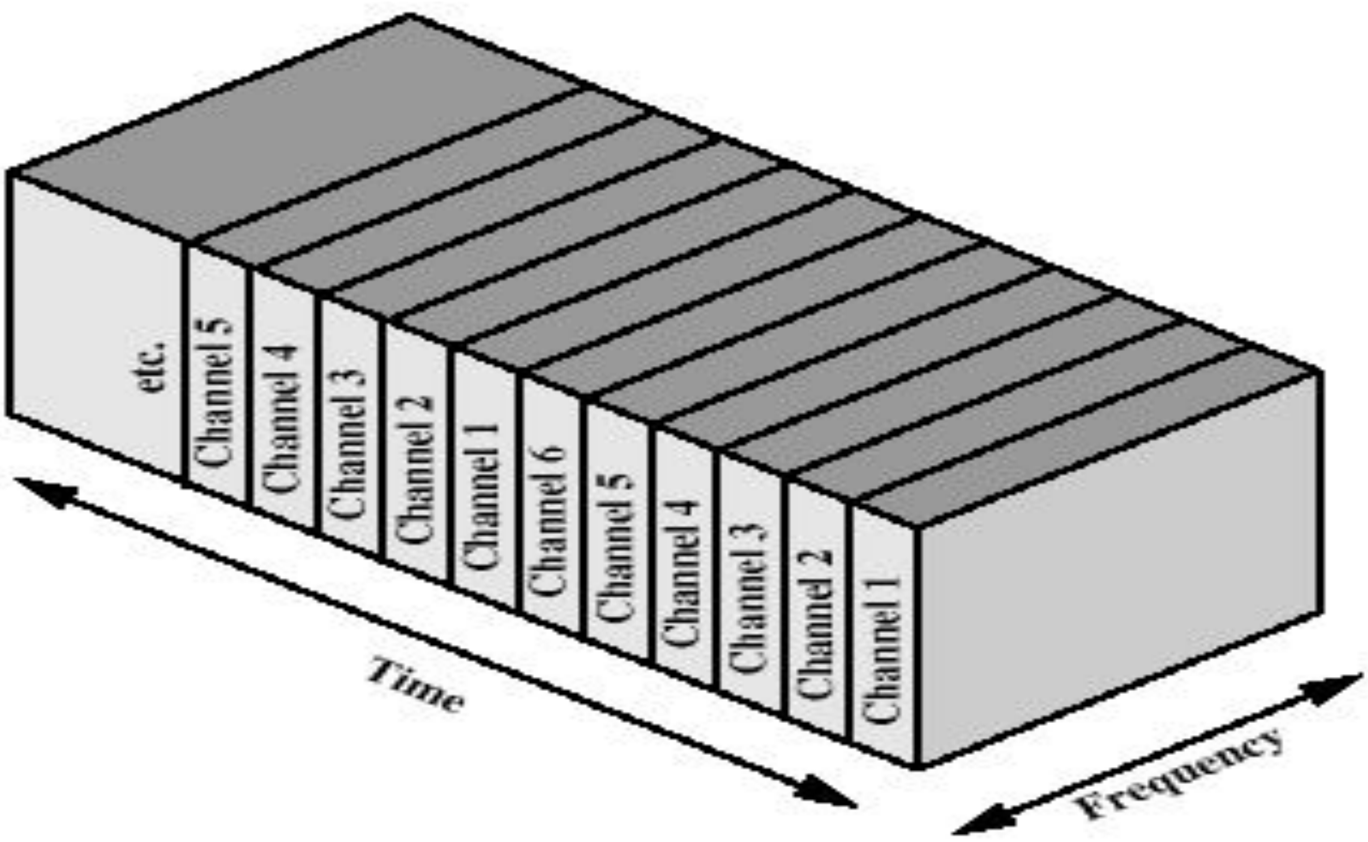
FDMA – Pros & Cons

- ✔ No dynamic coordination necessary
- ✔ Works well for analog signals
- ✔ Wastage of bandwidth in case of uneven traffic distribution
- ✔ Inflexible
- ✔ Need of guard spaces
- ✔ Higher costs of cell site system and mobile

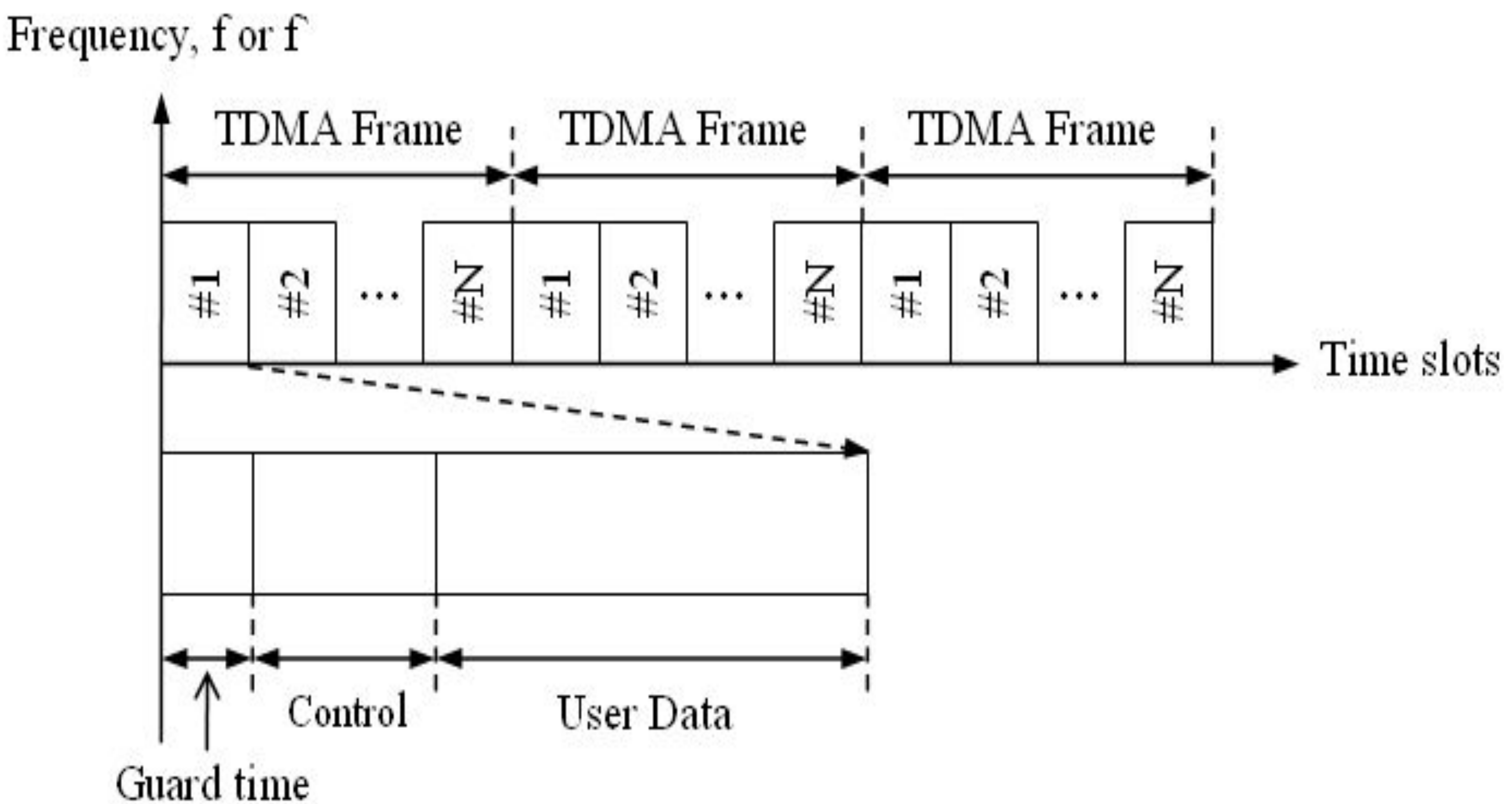
What is TDMA?

- TDMA refers to allowing a number of subscribers to access a specified channel bandwidth on a time-shared basis.
- It utilizes the digital technology with more efficient and complex strategies of sharing the available spectrum among number of subscribers simultaneously
- One carrier channel is used by several subscribers, and each subscriber is served in a round-robin method.

Time Division Multiple Access (TDMA)

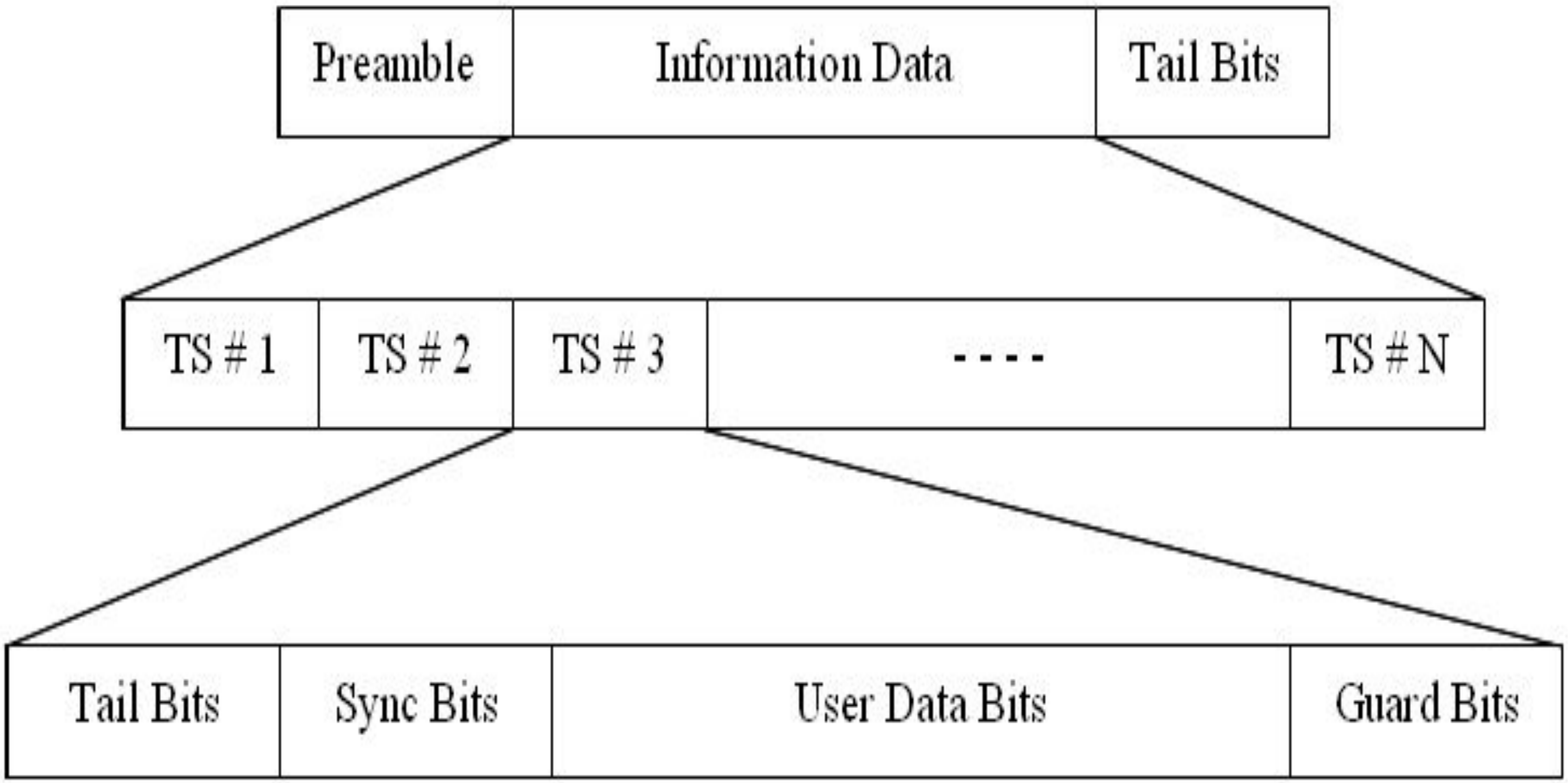


Typical Frame Structure of TDMA

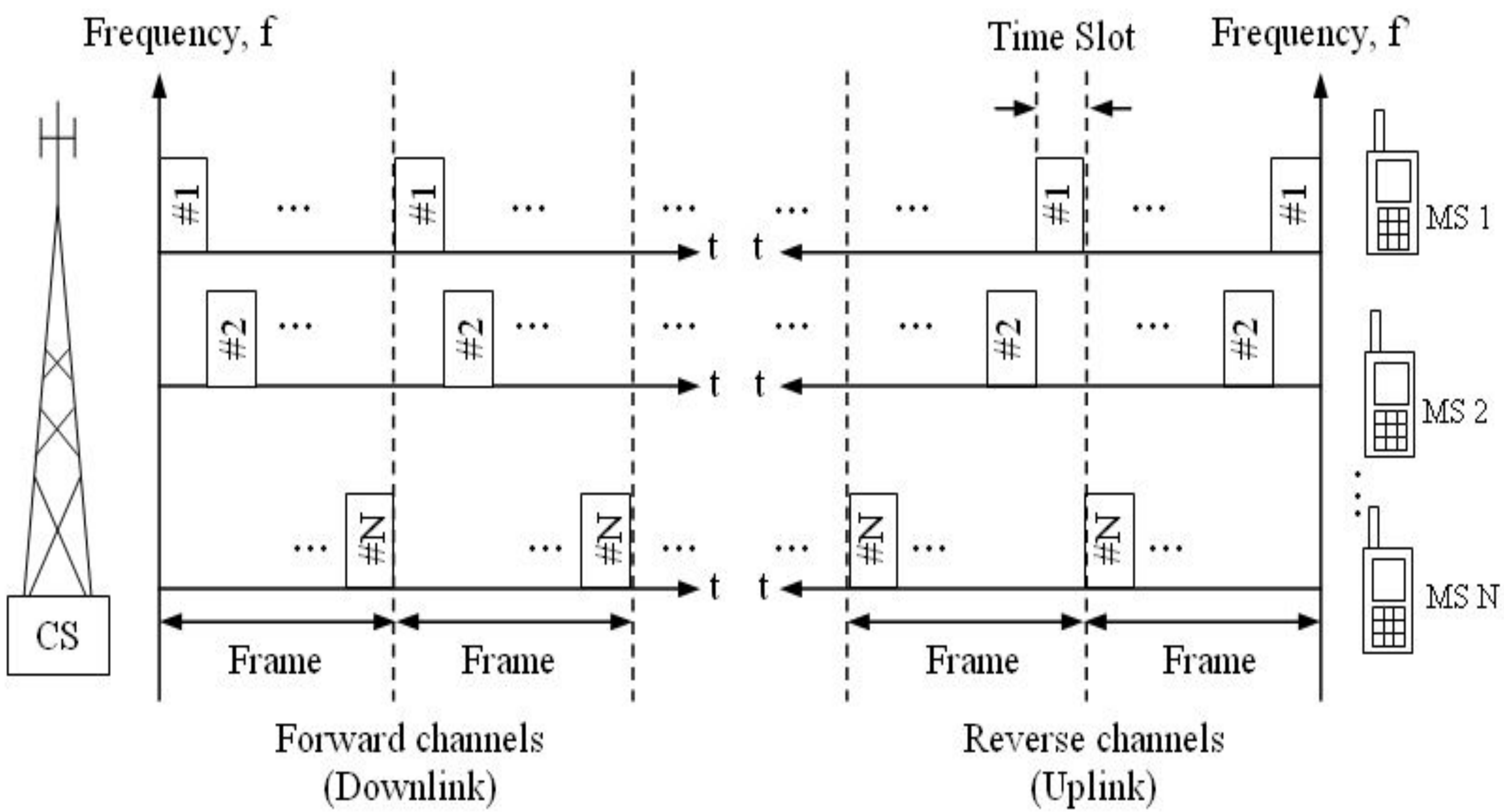


guard time between the slots so that interference due to propagation delays along different paths can be minimized

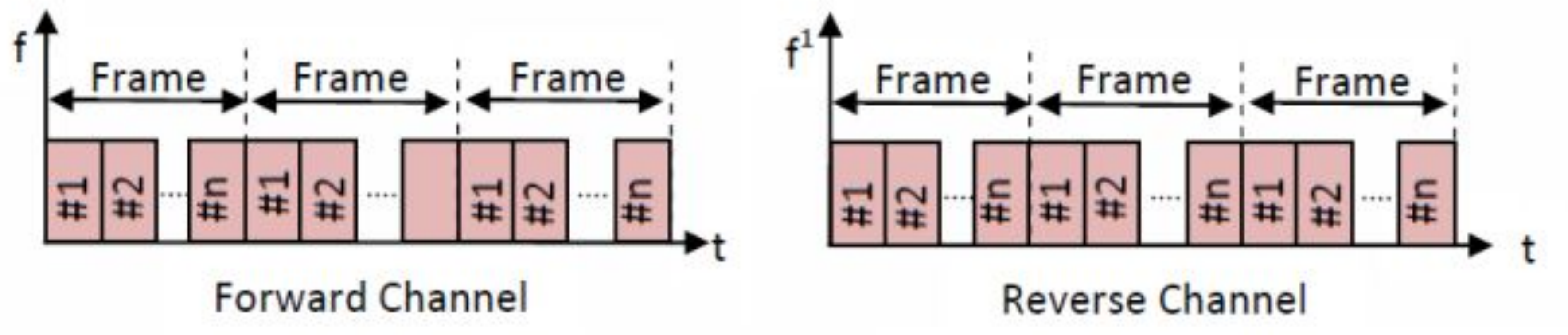
A TDMA Time Slot Structure



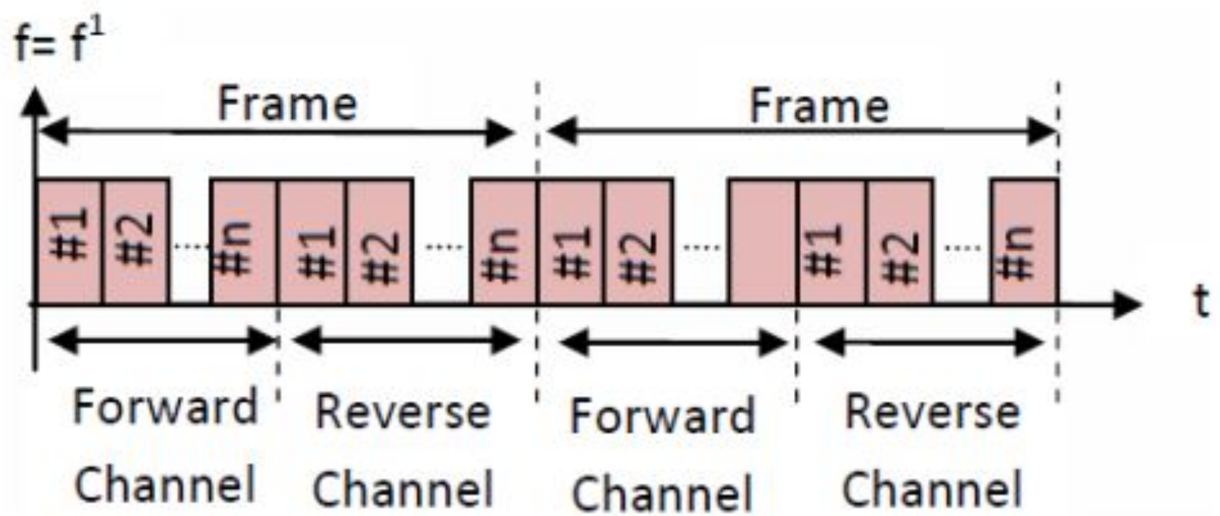
Basic Structure of TDMA System



TDMA – Forward and Reverse Channels

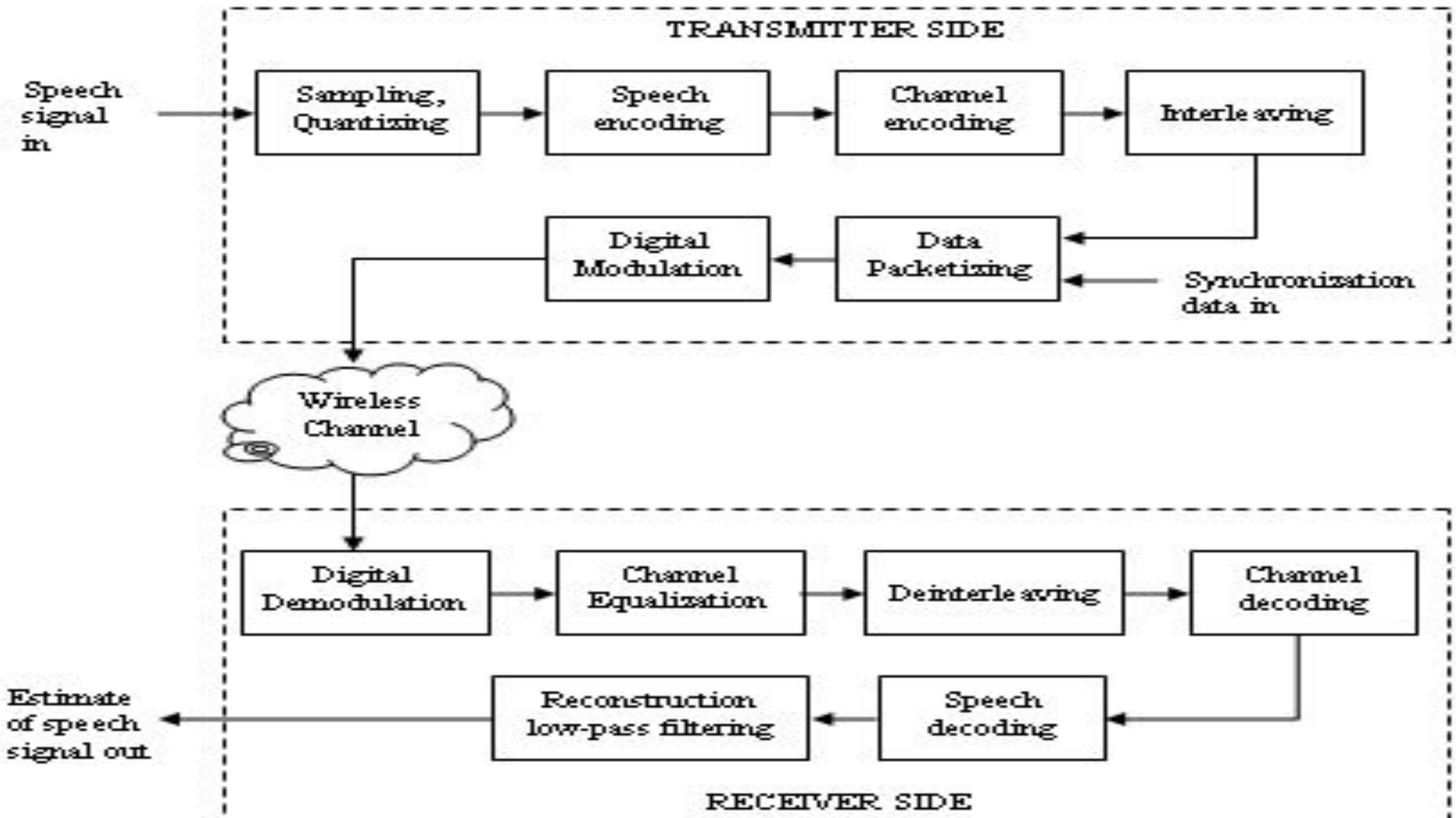


Structure of forward and reverse Channels in TDMA/FDD system



Structure of forward and reverse Channels in TDMA/TDD system

A Basic TDMA Link



The features of TDMA includes the following:

1. TDMA shares a single carrier frequency with several users where each user makes use of non overlapping time slots.
2. The number of time slots per frame depends on several factors such as modulation technique, available bandwidth.
3. Data transmission in TDMA is not continuous but occurs in bursts. This results in low battery consumption since the subscriber transmitter can be turned OFF when not in use. Because of a discontinuous transmission in TDMA the handoff process is much simpler for a subscriber unit, since it is able to listen to other base stations during idle time slots.
4. TDMA uses different time slots for transmission and reception thus duplexers are not required.
5. TDMA has an advantage that is possible to allocate different numbers of time slots per frame to different users.

Thus bandwidth can be supplied on demand to different users by concatenating or reassigning time slot based on priority.

Advantages:

1. It carries data rates of 64kbps to 120 Mbps
2. It provides the user with extended battery life and talk time.
3. TDMA technology separates users according to time; it ensures that there will be no interference.
4. TDMA allows the operator to do services like fax, voice bad data and SMS as well bandwidth intensive application such as multimedia and video conferencing.
5. Uplink power control is not required.
6. Transmission plans and capacity management is done by the satellite are very flexible

Disadvantages

1. Each user has predefined time slot. When moving from one cell to other, if all the time slots in this cell are full the user might be disconnected.
2. It is subjected to multipath distortion. A signal coming from a tower to a handset might come from any one of several directions. It might have bounced off several different buildings before arriving.
3. It requires a network wide time synchronization which makes the entire system very complex.
4. Analog of digital conversions are required.

TDMA – Pros & Cons

- ✔ Only one carrier in the medium at any time
- ✔ High throughput even for many users
- ✔ Accommodates transmission time delay
- ✔ Easy encryption and decryption of the signal
- ✔ Offers better signal quality in a mobile radio environment
- ✔ Precise synchronization necessary

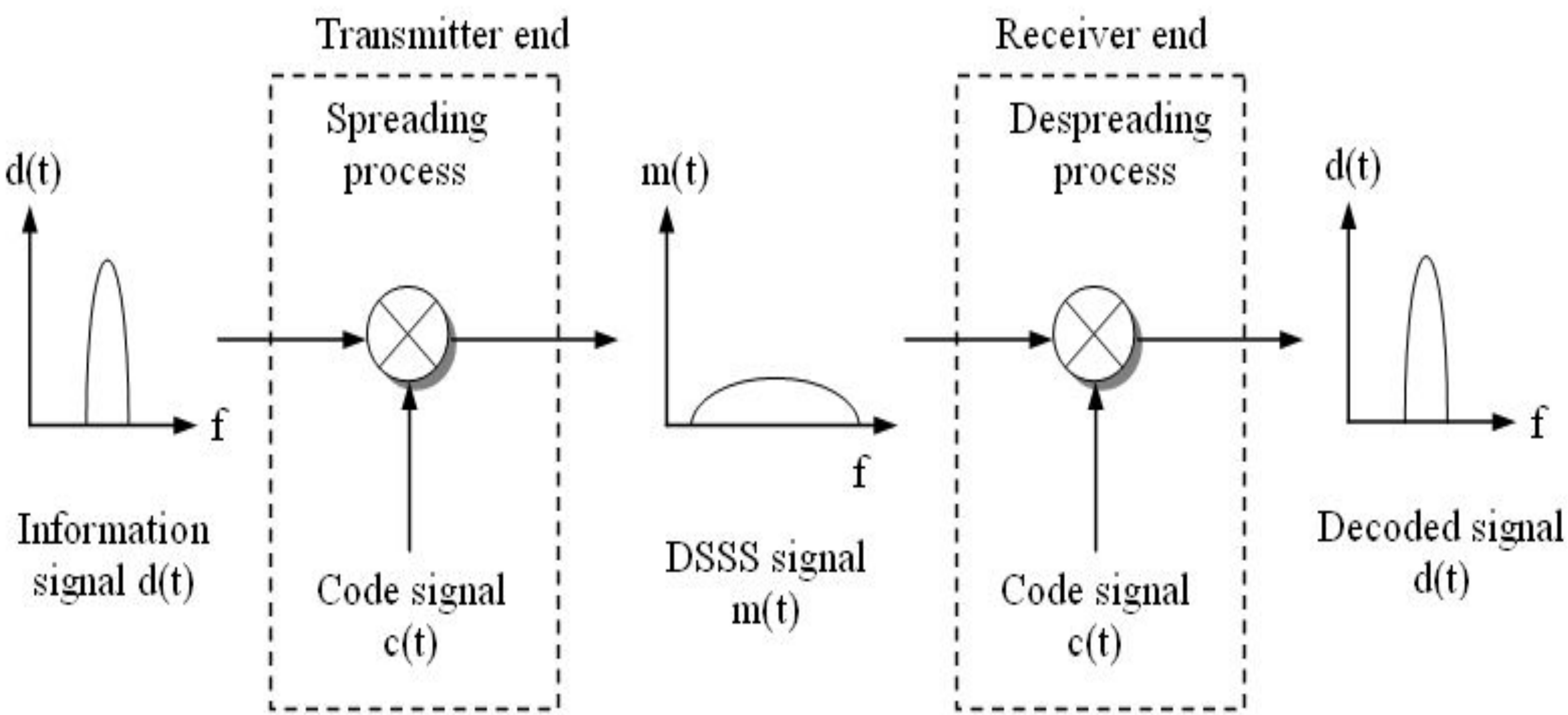
Spread Spectrum Multiple Access

- ✔ Spreads the information-bearing data signal over a large bandwidth
- ✔ Uses signals which have transmission bandwidth much greater than minimum required bandwidth.
- ✔ Allows same spectrum to be used simultaneously by many subscribers in adjacent cells
- ✔ Bandwidth efficient in a multiple subscriber environment

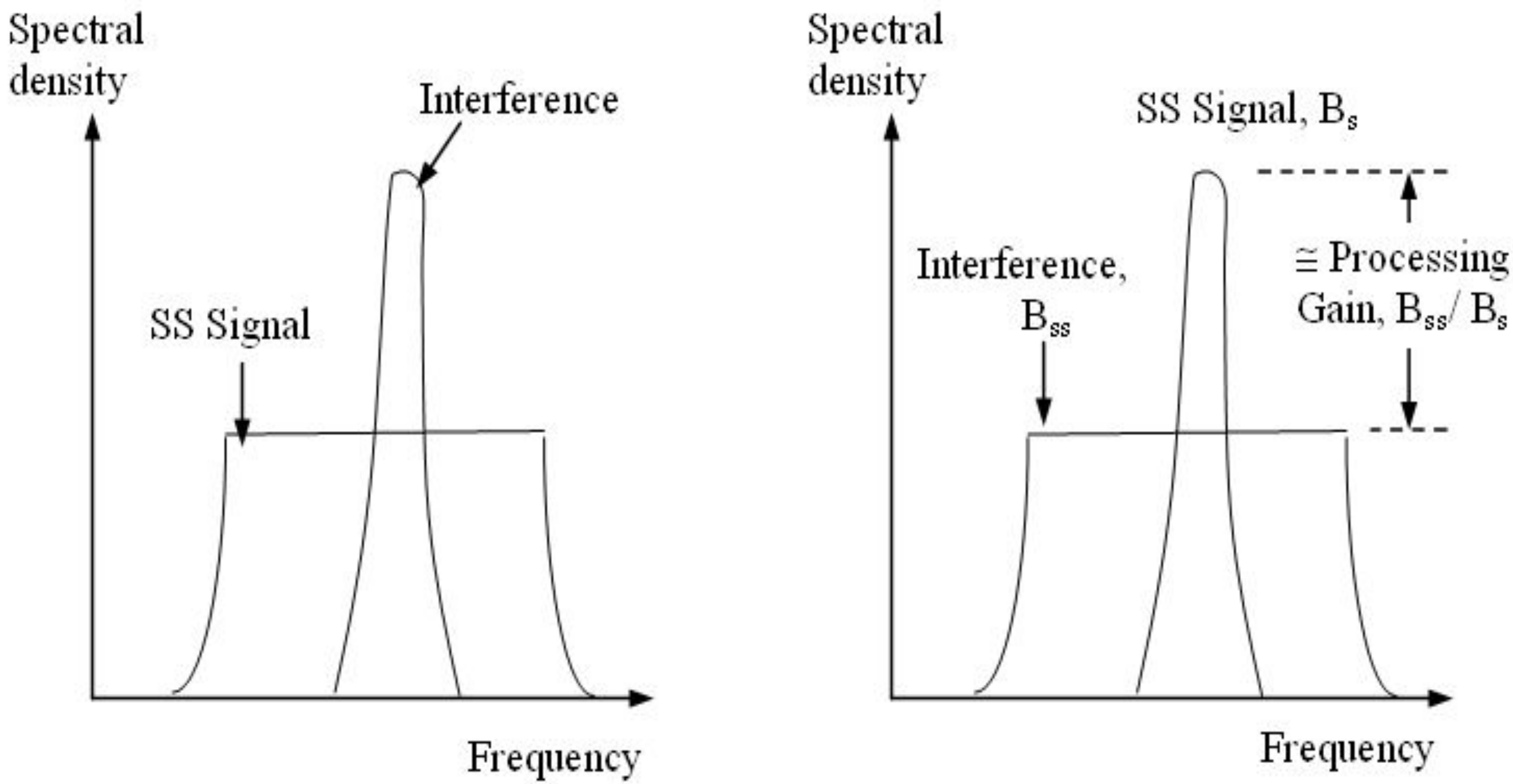
Criteria for Spread Spectrum Signal

- The bandwidth of the transmitted spread signal is much greater than the bandwidth of the information data signal.
- Orthogonal PN code sequence other than the information data being transmitted, determines the actual transmitted bandwidth on-the-air.

Basic Concept of Direct Sequence Spread Spectrum



Spectra of Received SS Signal

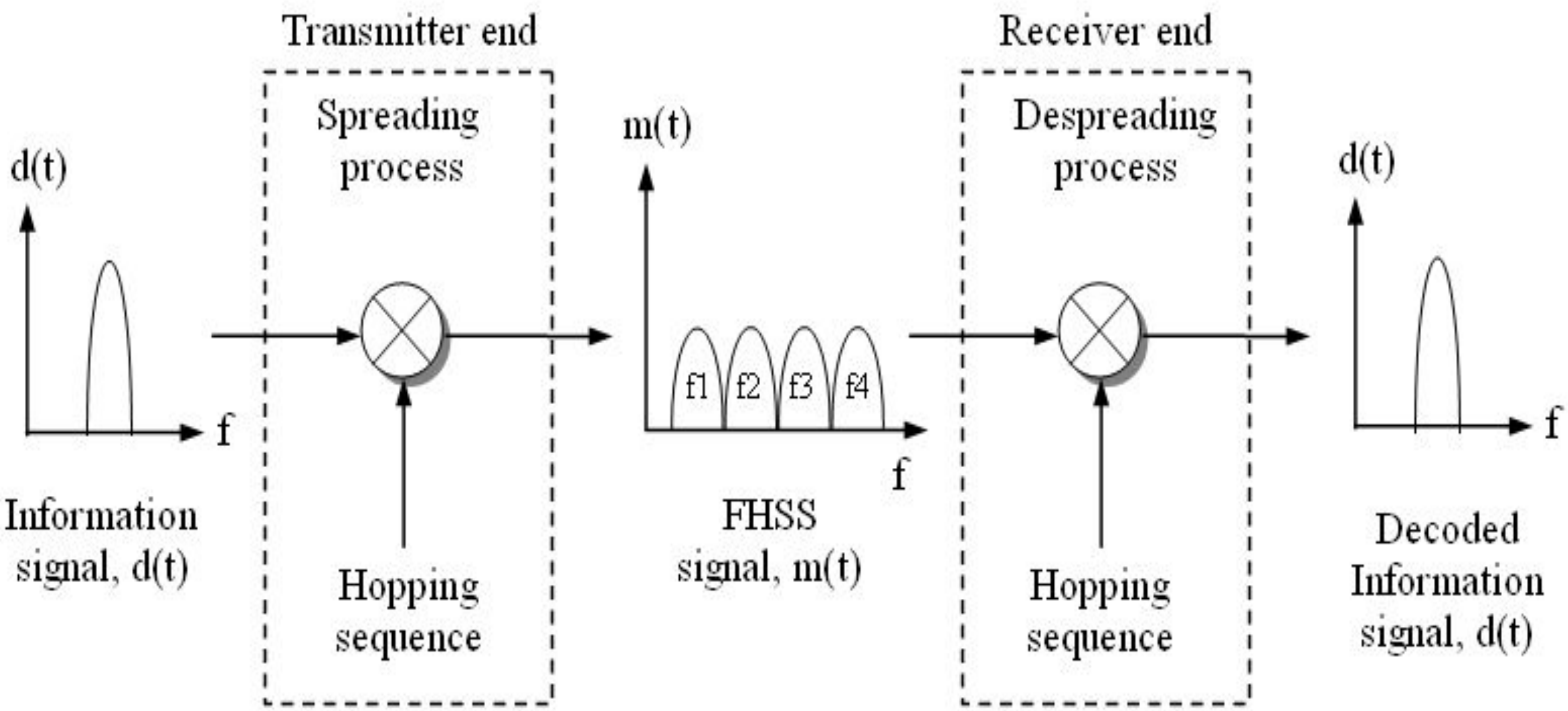


(a) correlator input before despreading (b) correlator input after despreading

Features of DSSS Technique

- Increased tolerance to interference
- Low probability of interception
- Increased tolerance to multipath interference
- Use of multipath energy to the advantage to improve the system performance
- Increased ranging capability

Basic Concept of Frequency Hopping Spread Sequence

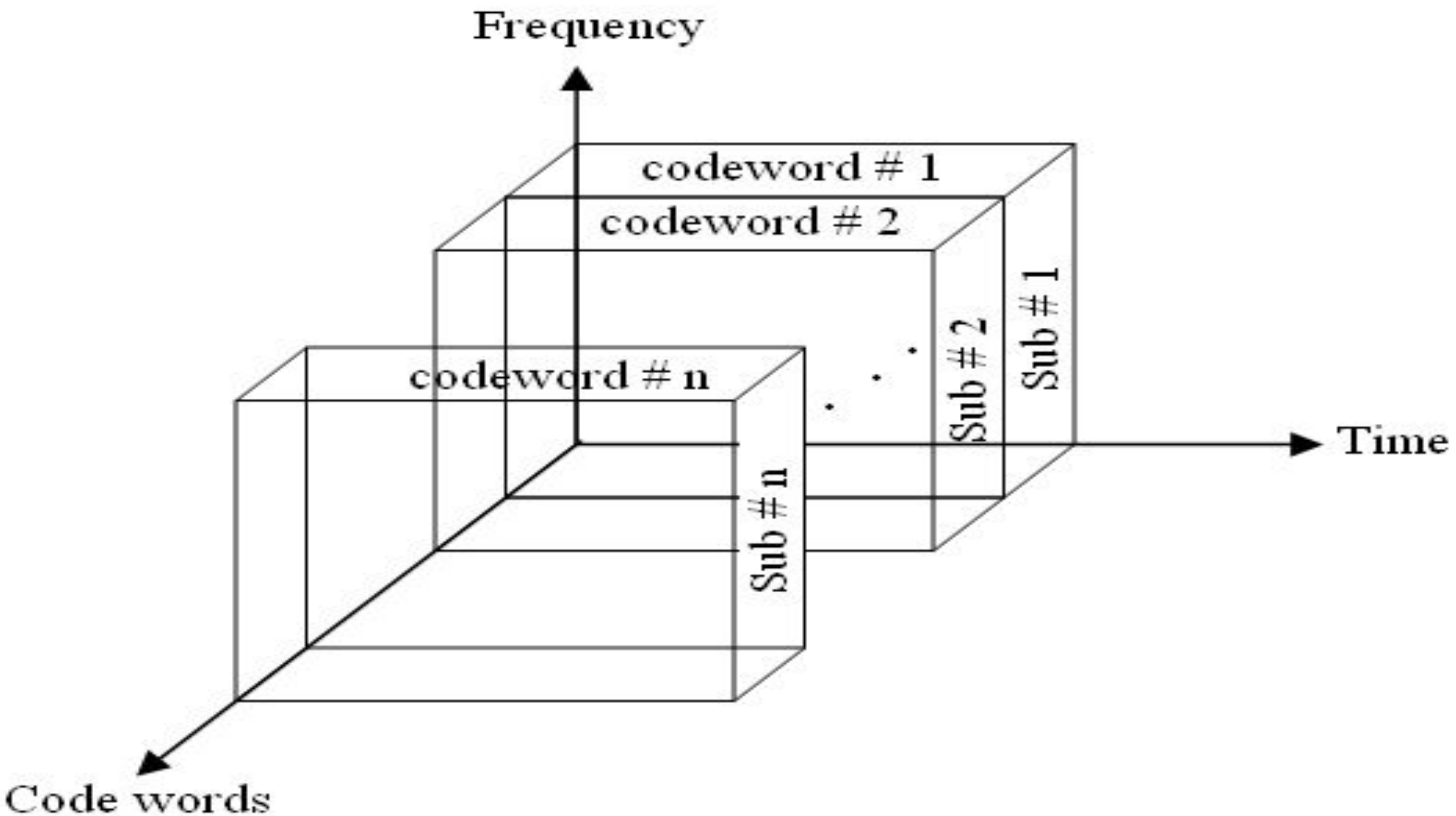


- In a FH method, a pseudorandom sequence is used to change the radio signal frequency across a broad frequency band in a random fashion.
- A spread spectrum modulation technique implies that the radio transmitter frequency hops from channel to channel in a predetermined but pseudorandom manner.
- The RF signal is dehopped at the receiver end using a frequency synthesizer controlled by a pseudorandom sequence generator synchronized to the transmitter's pseudorandom sequence generator.
- Multiple simultaneous transmission from several users is possible using FH, as long as each uses different frequency hopping sequences and none of them "collides" (no more than one unit using the same band) at any given instant of time.

Spread Spectrum and CDMA

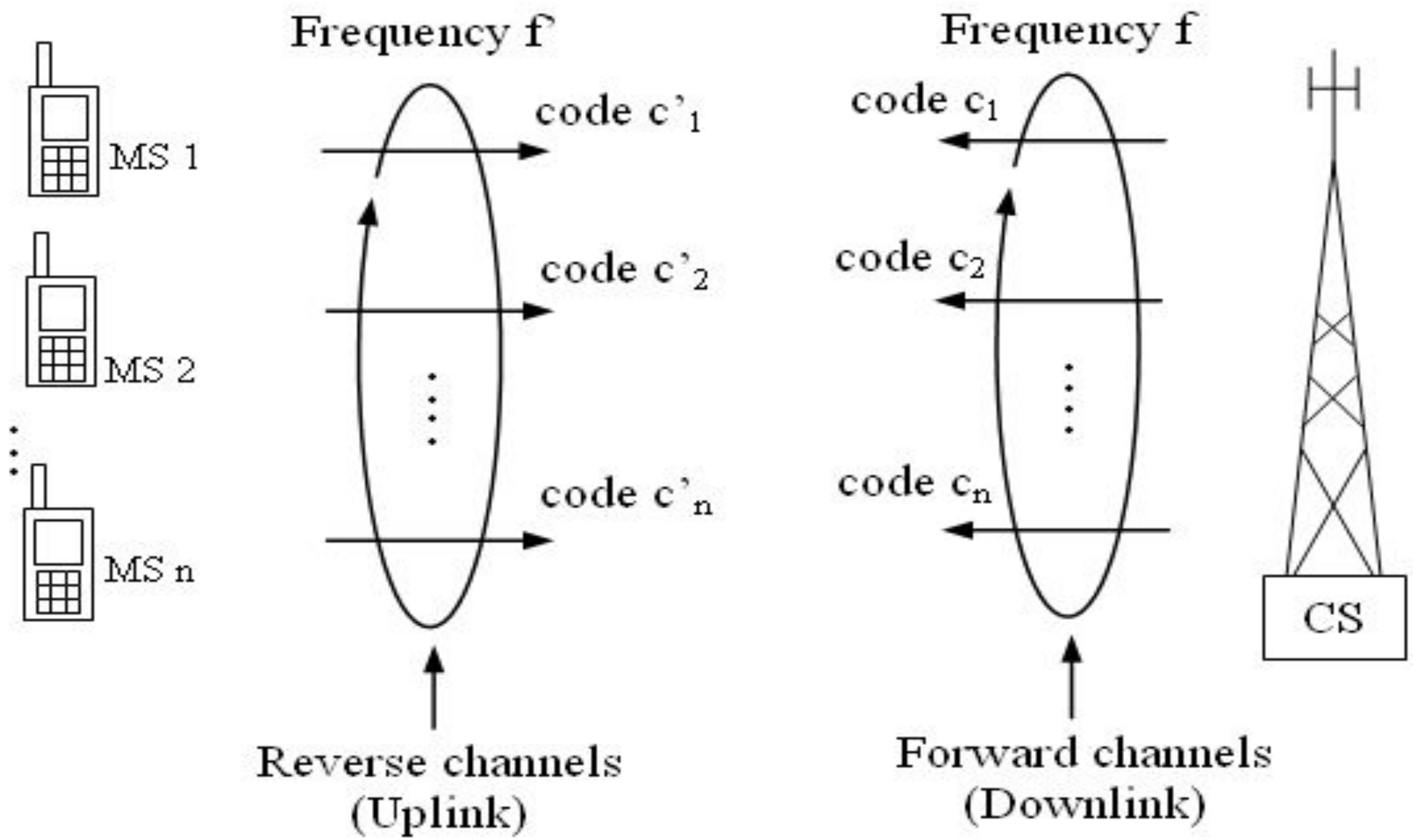
- Spread spectrum is a modulation technique
- It forms the basis for code-division multiple access (CDMA) technique
- In CDMA, individual mobile subscribers occupy the complete spectrum whenever they transmit.
- The signals are spread with a specific PN code to distinguish it from other signals

The Concept of CDMA



- In a CDMA system, different spread-spectrum codes are selected and assigned to each user, and multiple users share the same frequency.
- A CDMA system is based on spectrum-spread technology, which makes it less susceptible to the noise and interference by substantially spreading over the bandwidth range of the modulated signal.
- However, high efficiency of frequency usage has been demonstrated by using CDMA, since the introduction of power control enables us to adjust the antenna emitting power so that the near-far problem could be solved

Structure of a CDMA System



Advantage of CDMA

1. It gives good protection against interference and tapping.
2. Potentially larger capacity (more users can communicate simultaneously)
3. The transition from one Base station to another (handoff) is not abrupt as in TDMA and provides better quality.
4. Easy addition of more users.
5. Impossible for hackers to decipher the code sent.
6. Better signal quality.

Disadvantages:

1. As the number of users increases, the overall quality of services decreases.
2. Self jamming
3. Near-Far problem arises.
4. Relatively high complexity of the receiver. (A receiver has to know the code and must separate the channel with user data from the background noise composed of other signals and environmental noise. Additionally, a receiver must be precisely synchronized with the transmitter the decoding correctly.)
5. All signals should reach a receiver with almost equal strength; otherwise some signals could drain others. If some people close to a receiver talk very loudly the language does not matter. The receiver cannot listen to any other person. To apply CDM, precise power control is required

Multiple Access Techniques in Systems

S. No.	Type of Cellular System	Standard	Multiplexing Technique	Multiple Access Techniques
1.	1G Analog Cellular	AMPS	FDD	FDMA
2.	US Digital Cellular	USDC	FDD	TDMA
3.	2G Digital Cellular	GSM	FDD	TDMA
4.	Pacific Digital Cellular	PDC	FDD	TDMA
5.	US Narrowband Spread Spectrum Digital Cellular	IS-95	FDD	CDMA
6.	3G Digital Cellular	W-CDMA	FDD/TDD	CDMA
7.	3G Digital Cellular	Cdma2000	FDD/TDD	CDMA

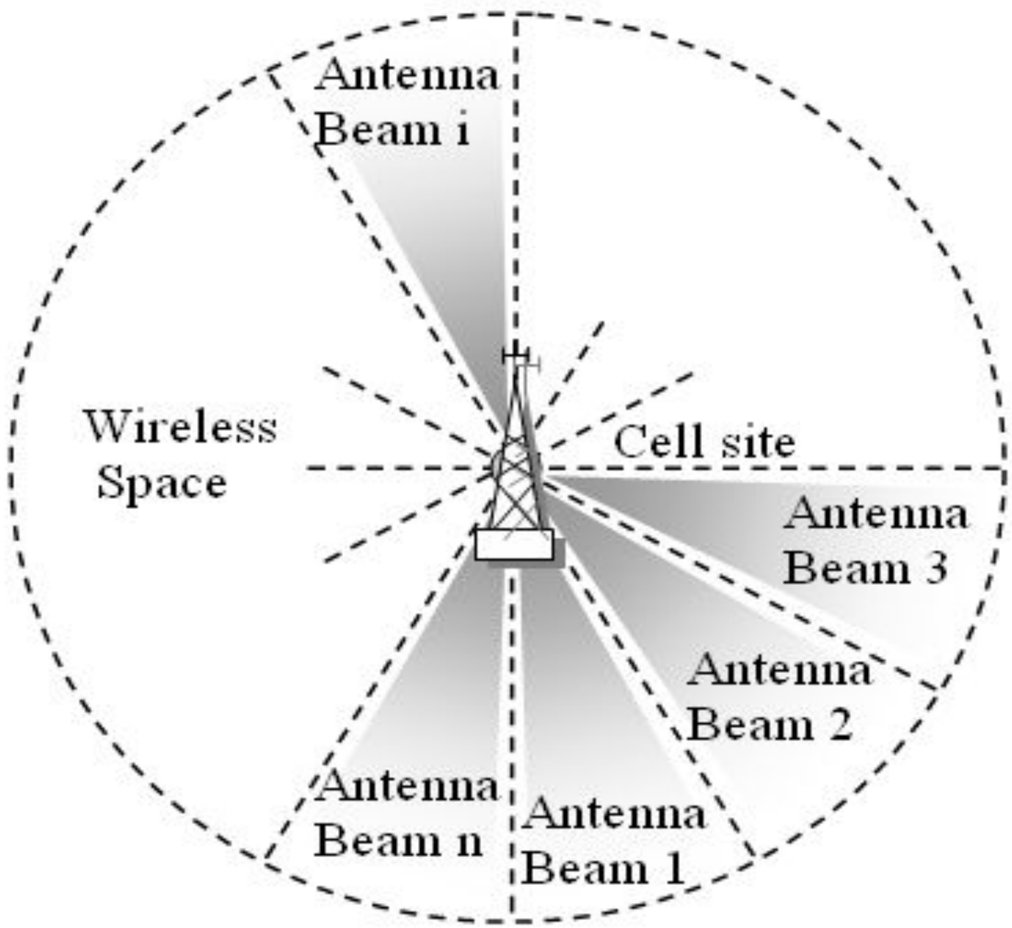
Comparison of Various Multiple Division Techniques

Technique	FDMA	TDMA	CDMA
Concept	Divide the frequency band into several subbands	Divide the time into non-overlapping time slots	Spread the signal with orthogonal codes
Active terminals	All terminals active on their specified frequencies	Terminals active on in their specified slot on same frequency	All terminals active on same frequency
Signal separation	Filtering in Frequency	Synchronization in time	Code separation
Handoff	Hard handoff	Hard handoff	Soft handoff
Advantages	Simple and robust	Flexible	Flexible
Disadvantages	Inflexible, available frequencies are fixed, requires guard bands	Requires guard space, synchronization problem.	Complex receivers, requires power control to avoid near-far problem

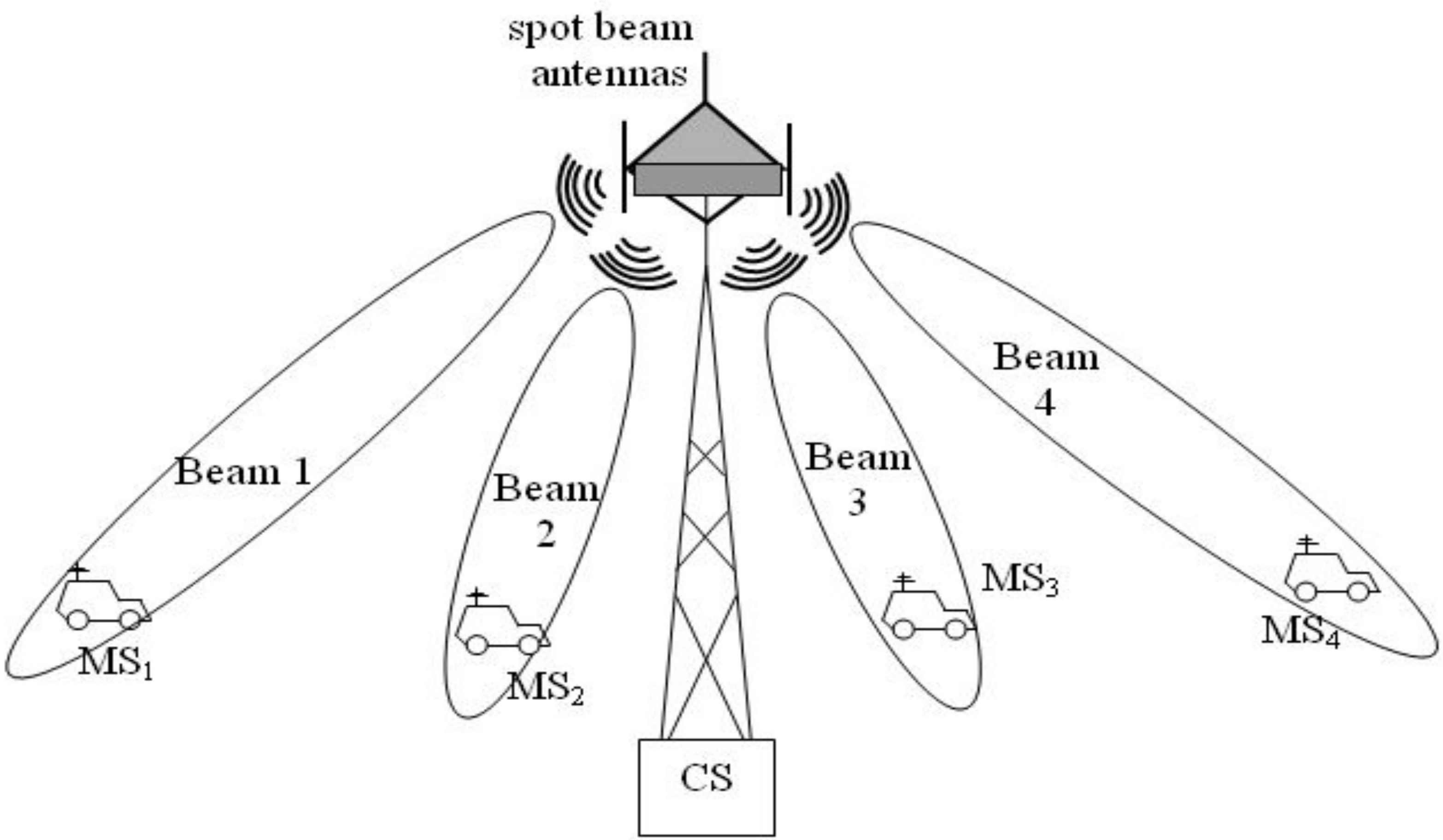
Space Division Multiple Access (SDMA)

- SDMA controls the radiated energy for each user in space by using spot beam antennas.
- Sectorized antennas and adaptive antennas are typical applications of SDMA.
- All users within the system are able to communicate at the same time using the same channel.
- Implemented using FDMA/TDMA/CDMA.

The Concept of SDMA



Basic Structure of SDMA System



Advantages of SDMA Approach

- Can be applied with FDMA, TDMA, or CDMA
- Allows many subscribers to operate on the same frequency and/or time slot in the same cell
- Leads to more number of subscribers within the same allocated frequency spectrum with enhanced user capacity
- Can be applied at the cell-site without affecting the mobile subscriber

SDMA and Smart Antennas

- The simplest example of a smart antenna is the use of **sector antennas** at the cell site.
- **Switched-beam antennas** have a number of fixed beams that cover 360°
- **Adaptive antennas** provide a dedicated beam for each subscriber
- Results in even greater reuse of the radio spectrum

Advantages of Smart Antennas

- Greater range due to directional antennas and larger gains
- Fewer cell sites
- Better signal penetration
- Less sensitivity to power control errors
- Responsive to hot spots traffic conditions

Hybrid Multiple Access Techniques

- Hybrid TDMA/FDMA
- Hybrid TDMA/DSMA
- Hybrid TDMA/FHMA
- Hybrid DSMA/FHMA
- Hybrid FDMA/DSMA
- Hybrid SDMA with FDMA/TDMA/ CDMA

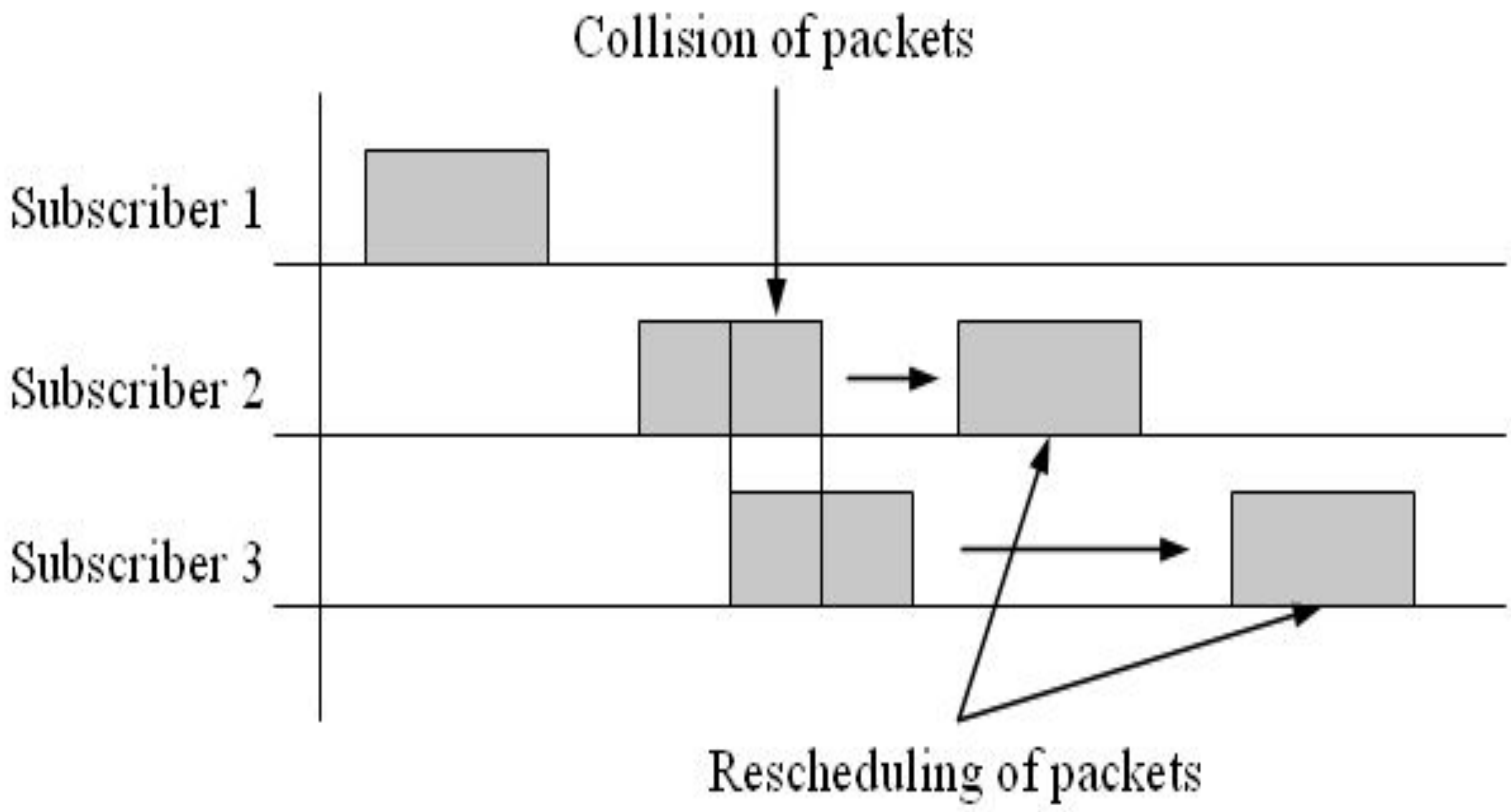
Packet Radio Multiple Access (PRMA) Techniques

- **Contention-based techniques can serve a large number of subscribers with least overhead or control mechanism.**
- The performance parameters are the average delay (D) experienced by a typical data packet and the throughput (T).
- Throughput is the average number of messages successfully transmitted per unit time.

Packet Radio Protocol : ALOHA

- Based on the type of access, contention protocols are categorized as random access, scheduled access, and hybrid access.
- The **pure ALOHA protocol** is a random access protocol used for data transfer.
- **Vulnerable period** is the time interval during which the packets are susceptible to collisions with transmissions from other users.

Concept of Pure ALOHA Protocol

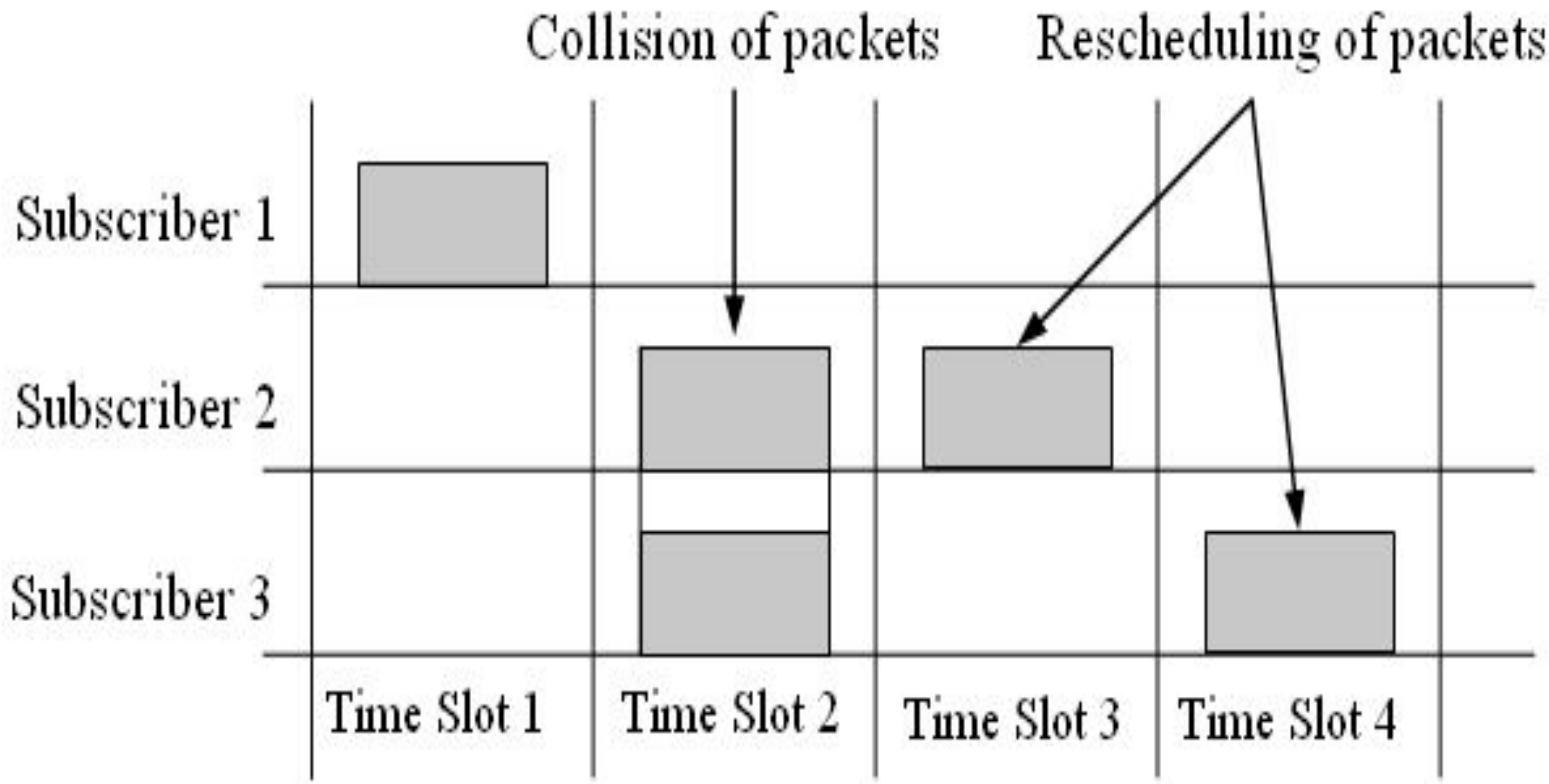


Throughput of Pure ALOHA

- **Throughput, S (ALOHA) = $\lambda\, e^{-2\lambda T}$**
- **Normalized S_0 (ALOHA) = $\lambda\, T\, e^{-2\lambda T}$**
- **S_0 (ALOHA) = $G\, e^{-2G}$**

  where **$G = \lambda T$** is the traffic occupancy or the normalized channel traffic (measured in Erlangs).
- **S_{0peak} (ALOHA) =0.184 packets per packet time for G = 0.5**

Concept of Slotted ALOHA Protocol

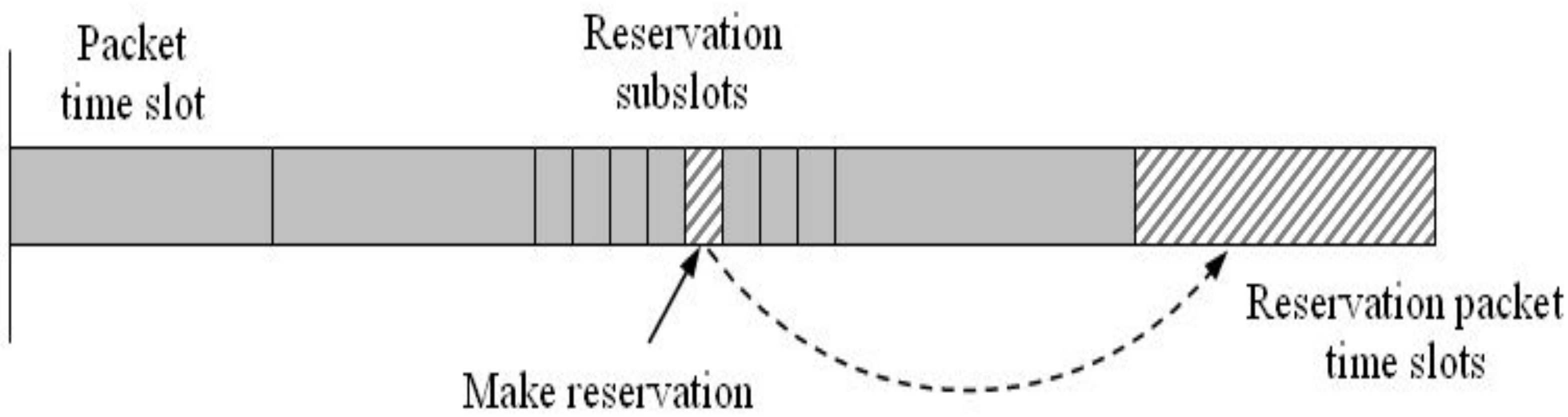


Throughput of Slotted ALOHA

- **Throughput, $S_S = \lambda e^{-\lambda T}$**
- **Normalized $S_{0S} = \lambda T e^{-\lambda T}$**
- **$S_{0S} = G e^{-G}$**

  where **$G = \lambda T$** is the traffic occupancy or the normalized channel traffic (measured in Erlangs).
- **S_{0peak} (Slotted ALOHA) =0.368 packets per packet time for G = 1**

Reservation ALOHA Protocol



- Reservation ALOHA (R-ALOHA) protocol is a combination of slotted ALOHA protocol with TDMA system.

Traffic Types versus Multiple Access

S. No.	Type of Traffic	Multiple Access Technique
1.	Voice	Contention-free multiple access techniques such as FDMA, TDMA, CDMA
2.	Bursty, short messages	Contention-based protocols such as Pure ALOHA, Slotted ALOHA
3.	Bursty, long messages, large number of subscribers	Reservation protocols – Reservation ALOHA
4.	Bursty, long messages, small number of subscribers	Reservation protocols with fixed TDMA reservation channel - PRMA

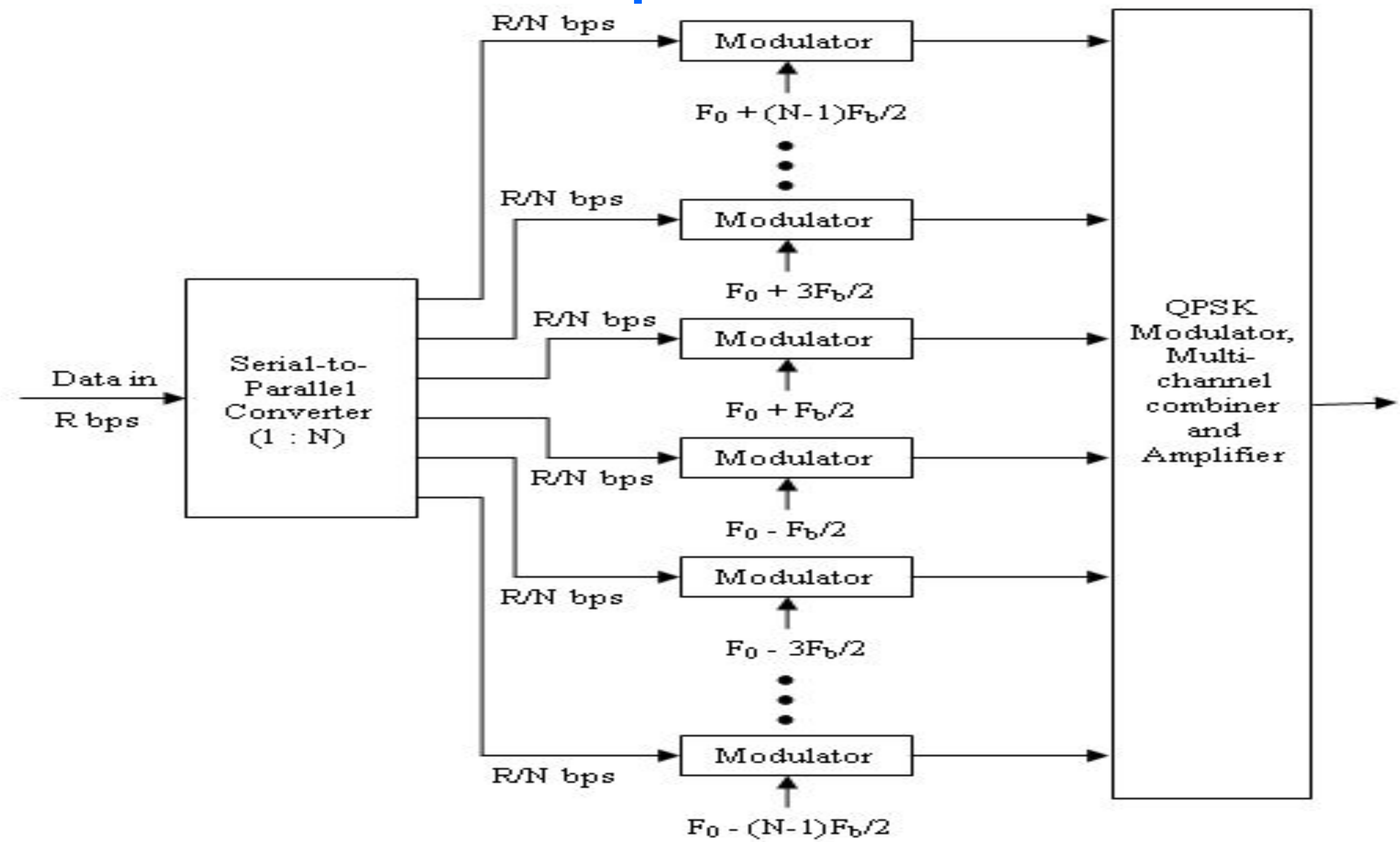
Carrier Sense Multiple Access Protocols

- Nonpersistent CSMA Protocol
- Persistent CSMA Protocol
- CSMA with Collision Detection (CSMA/CD)
- CSMA with Collision Avoidance (CSMA/CA)
  - **CSMA/CA with ACK**
  - **CSMA/CA with RTS and CTS**

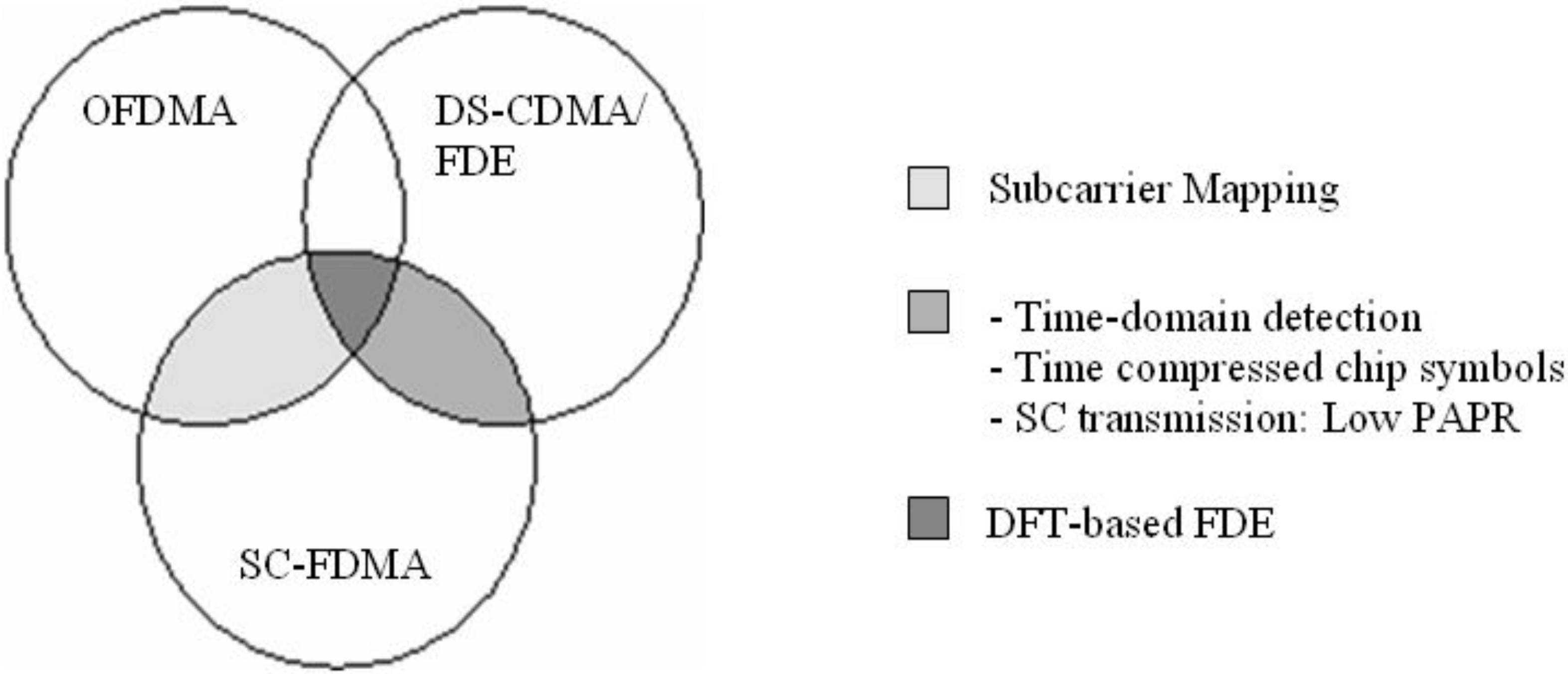
Multicarrier Multiple Access Schemes

- Orthogonal Frequency-Division Multiple Access (OFDMA)
- Single-carrier FDMA (SC-FDMA)
- Multi-Carrier Code Division Multiple Access (MC-CDMA)
- Multi-carrier direct sequence code division multiple access (MC-DS-CDMA)

Concept of OFDM



Relationship among Multi-carrier Multi-Access Schemes



Summary

- Multiple access techniques are efficient ways to utilize the limited radio spectrum by simultaneous users.
- **Number of subscribers need to access the control channel on shared basis at random times and for random periods.**
- Controlling access to a shared medium is important.

Thanks!